Check Points

Remark $G \in \mathbb{R}^{n\times n}$ $D \in \mathbb{R}^{n\times n}$ $L = D - G \in \mathbb{R}^{n\times n}$

Let N_i be the set of all neighbors of robot i

$d_i = \sum_{j=1}^{n} G_{ij} = \sum_{j\in N_i} G_{ij}$ $X \in \mathbb{R}^n$ $LX \in \mathbb{R}^n$

$$G_{ij} = \begin{cases} 1 & (j \in N_i) \\ 0 & (\text{else}) \end{cases} \quad D_{ij} = \begin{cases} d_i & (i=j) \\ 0 & (\text{else}) \end{cases} \quad L_{ij} = \begin{cases} d_i & (i=j) \\ -1 & (\text{else}) \end{cases}$$

$$(LX)_i = (DX)_i - (GX)_i = d_i x_i - \sum_{j\in N_i} 1 \times x_j$$

$$= \sum_{j\in N_i} x_i - \sum_{j\in N_i} x_j = \sum_{j\in N_i} (x_i - x_j) \quad \text{— Eq ①}$$

$W_A = (I + D)^{-1}(I + G)$, $W_A \in \mathbb{R}^{n\times n}$, $W_A X \in \mathbb{R}^n$

$W_B = (I - \varepsilon L)$, $W_B \in \mathbb{R}^{n\times n}$, $W_B x \in \mathbb{R}^n$

$$(W_A)_{ij} = \begin{cases} \frac{1}{1+d_i} & (i=j \text{ or } j \in N_i) \\ 0 & (\text{else}) \end{cases} \quad ②$$

$$(W_B)_{ij} = \begin{cases} 1 - \varepsilon d_i & (i=j) \\ \varepsilon & (j \in N_i, j \neq i) \\ 0 & (\text{else}) \end{cases} \quad ③$$

$$(W_A X)_i = \frac{x_i + \sum_{j\in N_i} x_j}{1 + d_i} \qquad (W_B X)_i = x_i - \varepsilon \sum_{j\in N_i} (x_i - x_j)$$

(a) $A: \; x[k+1] = W_A x[k]$

$$x_i[k+1] = (W_A X)_i[k] = \frac{x_i[k] + \sum_{j\in N_i} x_j}{1 + d_i}$$

$B: \; x[k+1] = W_B x[k]$

$$x_i[k+1] = (W_B X)_i[k] = x_i[k] - \varepsilon \sum_{j\in N_i} (x_i[k] - x_j[k])$$

(b) When $X = 1$, $x_i = x_j = 1$

$$(W_A X)_i = \frac{\sum_{j \in N_i} 1 + 1}{1 + d_i} = \frac{d_i + 1}{d_i + 1} = 1$$

$\therefore W_A 1 = 1$

If $x_i = x_j = c$, $X = c1$

$W_A X = c(W_A 1) = c1 = X$

So, when all robots are already in the same position, nothing will change anymore

(c) $V = (I + D)1 \quad G = G^T \quad V \in \mathbb{R}^n$

A: Let $d = (d_1, d_2 \cdots d_i \cdots d_n)^T = G1 = D1$

$$V^T W_A = (I + D) 1^T [(I+D)^{-1}(I + G)]$$

$$= 1^T (I + D)(I + D)^{-1}(I + G)$$

$$= 1^T (I + G) = 1^T + 1^T G = 1^T + (G1)^T$$

$$= 1^T + (D1)^T = (1(I + D))^T = V^T$$

$$x[k+1] = W_A x[k]$$

$$V^T X[k+1] = V^T W_A x[k] = V^T x[k]$$

$$\Rightarrow V^T X[k+2] = V^T x[k+1] W_A = V^T x[k+1]$$

$$\therefore V^T x[k] = V^T x[0]$$

$$\sum_{i=1}^{n} (1 + d_i) x_i[k] = \sum_{i=1}^{n} (1 + d_i) x_i[0]$$

Suppose at k step, $X = c1$, same position

$$c = \frac{\sum_{i=1}^{n} (1 + d_i) x_i[0]}{\sum_{i=1}^{n} (1 + d_i)}$$

which means that the final position is a weighted average of initial position, d_i determines the weigh.

B: $\mathbf{1}^T W_B = \mathbf{1}^T - \mathbf{1}^T \varepsilon L = \mathbf{1}^T - \varepsilon(\mathbf{1}^T D - \mathbf{1}^T G)$

$\because \mathbf{1}^T D = D\mathbf{1} = G\mathbf{1} = \mathbf{1}^T G$

$\therefore \mathbf{1}^T W_B = \mathbf{1}^T$

$X[k+1] = W_B X[k]$

$\mathbf{1}^T x[k+1] = \mathbf{1}^T W_B X[k] = \mathbf{1}^T X[k]$

$\mathbf{1}^T x[k+2] = \mathbf{1}^T W_B x[k+1] = \mathbf{1}^T x[k]$

$\Rightarrow \mathbf{1}^T X[k] = \mathbf{1}^T x[0]$

Suppose they finally go to same position c $\quad X[k] = c\mathbf{1}$

$nc = \sum_{i=1}^{n} x_i[0]$

$c = \frac{\sum_{i=1}^{n} x_i[0]}{n}$

So the final position is the average of initial positions of all robots.

(d) According to ③

$$(W_B)_{ij} = \begin{cases} 1 - \varepsilon d_i & (i = j) \\ \varepsilon & (j \in N_i, i \neq j) \\ 0 & (\text{else}) \end{cases}$$

if $\varepsilon = \frac{1}{\max_i d_i}$, $\varepsilon d_i < 1$

$\therefore 1 - \varepsilon d_i > 0$

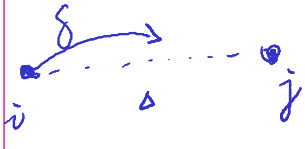
$\because d_i > 0$ for all i

$\therefore \varepsilon > 0$ $\therefore$ All entries ≥ 0

Recall: $X_i[k+1] = X_i[k] - \varepsilon \sum_{j \in N_i} (X_i[k] - X_j[k])$

Let $\Delta = X_i[k] - X_j[k]$, the distance between robot i and j in k^{th} step.

$\delta = X_i[k+1] - X_i[k]$, the change of position of X_i from step k to $k+1$



The purpose of using protocol A is to let Δ become smaller.

$\delta = -\varepsilon \Delta$

Let $X_i[k] - X_j[k] < 0$.

$\varepsilon = -\frac{\delta}{\Delta}$ $\quad \varepsilon > 0$

If $\varepsilon > 1$, $\delta > \Delta$

which means i will go over j, which is not what we expected